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#include <iostream>
#include <vector>
#include <cmath>
#include <iomanip> // Tambahkan ini untuk setprecision
using namespace std;

int main() {
    int n;
    cout << "Masukkan jumlah data (n): ";
    cin >> n;

    vector<double> x(n), y(n);
    cout << "Masukkan data X dipisah spasi:\n";
    for (int i = 0; i < n; i++) cin >> x[i];

    cout << "Masukkan data Y dipisah spasi:\n";
    for (int i = 0; i < n; i++) cin >> y[i];

    double sum_x = 0, sum_y = 0, sum_x2 = 0, sum_y2 = 0, sum_xy = 0;

    for (int i = 0; i < n; i++) {
        sum_x += x[i];
        sum_y += y[i];
        sum_x2 += x[i] * x[i];
        sum_y2 += y[i] * y[i];
        sum_xy += x[i] * y[i];
    }

    double x_mean = sum_x / n;
    double y_mean = sum_y / n;
```

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double ssx = sum_x2 - n * x_mean * x_mean;
double ssy = sum_y2 - n * y_mean * y_mean;
double ssxy = sum_xy - n * x_mean * y_mean;
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double b = ssxy / ssx;
double a = y_mean - b * x_mean;
double r = ssxy / sqrt(ssx * ssy);
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cout << fixed << setprecision(3);
cout << "\n=== Hasil Regresi Linear ===\n";
cout << "Rata-rata X: " << x_mean << endl;
cout << "Rata-rata Y: " << y_mean << endl;
cout << "SSX: " << ssx << endl;
cout << "SSY: " << ssy << endl;
cout << "SSXY: " << ssxy << endl;
```

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cout << setprecision(2); // khusus untuk r
cout << "Korelasi Pearson (r): " << r << endl;
```

```
cout << setprecision(3); // balik lagi ke 3 untuk a, b, dan prediksi
cout << "Nilai a (intersep): " << a << endl;
cout << "Nilai b (kemiringan): " << b << endl;
cout << "Persamaan regresi: y = " << a << " + " << b << "x" << endl;
```

```
char pred;
cout << "\nIngin memprediksi nilai Y berdasarkan X? (y/n): ";
cin >> pred;
```

```
while (pred == 'y' || pred == 'Y') {
    double x_pred;
    cout << "Masukkan nilai X yang ingin diprediksi: ";
    cin >> x_pred;
```

```
double y_pred = a + b * x_pred;  
cout << fixed << setprecision(3);  
cout << "Hasil prediksi Y: " << y_pred << endl;  
cout << "Prediksi lagi? (y/n): ";  
cin >> pred;  
}
```

```
cout << "Selesai. Terima kasih!" << endl;  
return 0;  
}
```